



Urban Informatics
Assignment - Week 8

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 13

Total marks: (6 X 1) + (7 X 2) = 20 marks

MCQ/MSQ Question

QUESTION 1: [2 marks]

_____ and _____ are supervised learning classification models.

- a. Logistic regression, multinomial logistic regression
- b. K-Means Clustering and DBSCAN
- c. PCA and autoencoders
- d. Apriori and decision tree

Correct Answer: a. Logistic regression, multinomial logistic regression

Detailed explanation: Logistic regression and multinomial logistic regression are supervised learning classification models.
Refer to lecture 36, slide 3.

QUESTION 2: [1 mark]

Which method is commonly used to estimate model parameters by maximizing the probability of the observed choices?

- a. Moving Average Method
- b. Monte Carlo Simulation
- c. Maximum Likelihood method
- d. Least Squares Method

Correct Answer: c. Maximum Likelihood method

Detailed explanation: The maximum likelihood method is used to estimate the parameters by maximizing the likelihood of the observed choices as per the model.
Refer to lecture 36, slide 7.



QUESTION 3: [2 marks]

Which of the following assumptions are CORRECT for Multinomial Logistic Regression?

- P. Observations are independent
- Q. Log-odds are non-linear in predictors
- R. Perfect collinearity among predictors
- S. Sufficient sample size for all classes

- a. P and S
- b. Q and R
- c. Q and S
- d. P and R

Correct Answer: a. P and S

Detailed explanation: Assumptions for Multinomial Logistic Regression are observations are independent, log-odds are linear in predictors, no perfect multicollinearity among predictors, IIA assumption holds, sufficient sample size for all classes and no complete separation in data (No predictor perfectly predicts a class).
Refer to lecture 37, slide 3.

QUESTION 4: [2 marks]

Match the following distance metrics with their description -

Distance metrics	Description
P. Cosine Distance	i. Measures distance along axes (grid-based movement for grid based layouts)
Q. Hamming Distance	ii. Measures orientation and ignores absolute size differences
R. Mahalanobis Distance	iii. Measures the number of mismatched categorical attributes between two observations
S. Manhattan (City-Block) Distance	iv. Used when features are correlated

- a. P-i; Q-iii; R-ii; S-iv
- b. P-ii; Q-iii; R-iv; S-i
- c. P-ii; Q-iv; R-ii; S-iii
- d. P-iv; Q-ii; R-i; S-iii

Correct Answer: b. P-ii; Q-iii; R-iv; S-i

Detailed explanation: Refer to lecture 38, slide 4 and 5.

QUESTION 5: [1 mark]

Support vectors in SVM are:

- a. misclassified data points only
- b. randomly selected data points
- c. data points farthest to the decision boundary
- d. data points closest to the decision boundary



Correct Answer: d. data points closest to the decision boundary

Detailed explanation: Support vectors are the data points closest to the decision boundary. Refer to lecture 39, slide 4.

QUESTION 6: [1 mark]

What is the main purpose of Soft Margin Support Vector Machines (SVM)?

- a. To completely eliminate classification errors
- b. To allow some margin violations and misclassifications for noisy data.
- c. To use only perfectly separable data
- d. To reduce the number of features in the dataset

Correct Answer: b. To allow some margin violations and misclassifications for noisy data.

Detailed Solution: Soft Margin SVM allows some training points to violate the margin or be misclassified in order to handle noisy, overlapping real-world data. Refer to lecture 39, slide 6.

QUESTION 7: [2 marks]

Match the following kernels with their description -

Kernels	Description
P. Linear kernel	i. Captures feature interactions with moderate complexity by modeling nonlinear relationships explicitly.
Q. Polynomial kernel	ii. Most commonly used kernel that models local similarity and creates highly flexible nonlinear decision boundaries.
R. RBF (Gaussian) kernel	iii. Mimics neural network activation behavior and is less used due to instability and tuning difficulties.
S. Sigmoid kernel	iv. Best suited for high-dimensional, sparse data such as text features or indicator variables.

- a. P-iv; Q-iii; R-ii; S-i
- b. P-ii; Q-iii; R-iv; S-i
- c. P-iii; Q-ii; R-i; S-iv
- d. P-iv; Q-i; R-ii; S-iii

Correct Answer: d. P-iv; Q-i; R-ii; S-iii

Detailed Solution: Refer to lecture 39, slide 8.

QUESTION 8: [2 marks]

SVM optimization is _____ and _____, ensuring a single global optimum for both hard and soft margin formulations.

- a. convex and quadratic
- b. concave and linear
- c. convex and linear
- d. concave and quadratic

Correct Answer: a. convex and quadratic



Detailed Solution: SVM optimization is convex and quadratic, ensuring a single global optimum for both hard and soft margin formulations.
Refer to lecture 39, slide 9.

QUESTION 9: [1 mark]

The model that learns jointly from both labeled and unlabeled (usually larger) data.

- a. Reinforcement learning
- b. Supervised learning
- c. Semi-supervised learning
- d. Association learning

Correct Answer: c. Semi-supervised learning

Detailed Solution: Refer to lecture 40, slide 3.

QUESTION 10: [2 marks]

Which of the following are assumptions of semi-supervised learning?

- P. Smoothness assumption
 - Q. Cluster assumption
 - R. Manifold assumption
 - S. Multicollinearity assumption
- a. P, Q and S
 - b. P, Q and R
 - c. Q, R and S
 - d. P, R and S

Correct Answer: b. P, Q and R

Detailed Solution: Refer to lecture 40, slide 4.

QUESTION 11: [1 mark]

Which statement best describes the Label Propagation Algorithm?

- a. It uses only labeled data and ignores unlabeled data.
- b. It assigns labels to data points without using similarity information
- c. It removes unlabeled data before classification.
- d. It iteratively spreads labels from labeled nodes to unlabeled nodes based on data similarity

Correct Answer: d. It iteratively spreads labels from labeled nodes to unlabeled nodes based on data similarity

Detailed Solution: Refer to lecture 40, slide 7.



QUESTION 12: [2 marks]

Match the following sampling methods in Active learning and Semi-Supervised learning with their descriptions -

Sampling methods	Description
P. Uncertainty sampling	i. Select points closest to the decision boundary
Q. Margin sampling	ii. Select samples where the model shows low confidence predictions
R. Entropy-based sampling	iii. Select samples that best represent different regions of the data.
S. Diversity sampling	iv. Choose samples with highest prediction entropy

- a. P-ii; Q-iii; R-i; S-iv
- b. P-iv; Q-iii; R-ii; S-i
- c. P-ii; Q-i; R-iv; S-iii
- d. P-iii; Q-ii; R-i; S-iv

Correct Answer: c. P-ii; Q-i; R-iv; S-iii

Detailed Solution: Refer to lecture 40, slide 12.

QUESTION 13: [1 mark]

In cluster-based semi-supervised learning, in which type of cluster assignment are unlabeled data points assigned the most frequent label?

- a. Majority voting assignment
- b. Centroid-based assignment
- c. Distance-based assignment
- d. Soft assignment

Correct Answer: a. Majority voting assignment

Detailed Solution: Refer to lecture 40, slide 11.

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